

Individual Reflection

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Course: AIG 130 - Cloud Computing Using Machine Learning

1. Task Distribution and My Specific Role

For this project, I acted as a General Assistant for my group. Instead of just focusing on one small part, this role allowed me to see the "big picture" of how a machine learning pipeline works, from loading the Yelp dataset to comparing the final results of different models.

2. Challenges and Key Learnings

The biggest challenge was understanding the technical trade-offs between the two models we used. I realized that when we used LightGBM, we had to turn sentences into vectors using TF-IDF. However, I noticed a major problem: this method treats a sentence like a "bag of words," which means we lose all the position and order information of the words.

I suspect this is why the Transformer-based model (DistilBERT) performed better. It keeps all the encoding information and understands how words relate to each other in a specific order. Even though deep learning takes more time and computer power, it is much "smarter" at reading human emotions because it doesn't lose that context.

3. Advantages and Disadvantages of Using Cloud Services

Our experience with Google Colab showed us exactly why cloud services are essential today:

- **Advantages:**

- Efficiency through Transfer Learning: This was my first time using Transfer Learning. By using the cloud, we could take a pre-trained DistilBERT model, freeze the first few layers, and only train the last layer. This saved us a huge amount of time compared to training a model from zero in the local environment.
- High-Performance Hardware: We didn't need expensive computers because the cloud provided GPUs for free. This made the deep learning part possible for us as students.

- **Disadvantages:**

- Resource Consumption: Even with transfer learning, deep learning is "heavy." It uses much more memory and energy than simple models like LightGBM.
- Connectivity: We were completely dependent on Google's servers. If the free Colab session timed out, we had to restart the environment and reload the datasets, which could be frustrating.